

MATÍAS FERNÁNDEZ LAKATOS

Machine Learning Engineer · Senior Neuromorphic Researcher · PhD

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· [ORCID](#) · [Portfolio](#)

PROFESSIONAL SUMMARY

Machine learning researcher and engineer with 10+ years of experience in statistical modeling and scientific computing, currently **Senior Neuromorphic Researcher at Gradiant**, researching third-generation neural networks (Spiking Neural Networks) and energy-efficient AI for edge deployments. Previously designed, trained, and deployed **unsupervised anomaly detection models** (β -VAE, autoencoders, isolation forests) for real-time cybersecurity on large-scale data streams, with end-to-end ownership: custom architectures in TensorFlow/Keras, hyperparameter optimization, MLflow experiment tracking, threshold calibration, and explainability (XAI). PhD-level foundation in optimization, multivariate statistics, and signal processing.

PROFESSIONAL EXPERIENCE

Senior Neuromorphic Researcher

Gradiant · Vigo, Spain · *Jul 2026 – present*

- Leading research on **Spiking Neural Networks (SNNs)** — third-generation neural networks with biologically inspired neuron models and **synaptic plasticity**-based learning — and their application to different data types (images, time series).
- Investigating the massive parallelism and **energy efficiency** of neuromorphic systems for **AI-on-the-edge** use cases.
- Researching frameworks for **compiling SNN models onto FPGAs** and neuromorphic-specific hardware, including links to Gradiant's photonic architectures.
- Investigating novel **explainable AI (xAI)** techniques for SNN-based models.
- Researching **neuromorphic vision sensors (DVS)** and event-based data processing; mapping applications in autonomous vehicles, manufacturing, and healthcare.
- Maintaining the **state of the art**: analysis of new publications, hardware devices, and software tools, contextualized against project objectives.

Research Engineer (Machine Learning)

Gradiant · Vigo, Spain · *2025 – Jun 2026*

- Designed and implemented unsupervised anomaly detection models (**β -VAE, Autoencoder, Isolation Forest, Extended Isolation Forest**) for early threat detection in corporate cybersecurity, building custom VAE architectures from scratch in **TensorFlow/Keras**.
- Led training strategy, evaluation framework, and **hyperparameter optimization** (grid/random search), with **MLflow** for experiment tracking, artifact logging, and model versioning.
- Calibrated anomaly detection thresholds via percentile-based selection, optimizing decision boundaries with **G-mean and loss-based criteria**.
- Evaluated models under **streaming constraints (~300K events)**, balancing detection performance with training/prediction time; resolved memory bottlenecks with batch-based processing.
- Applied **XAI techniques (SHAP)** so security analysts and non-technical users could interpret model decisions.
- Integrated models into a scalable architecture (**Kafka, Spark, Airflow, MinIO**); authored technical documentation for stakeholders (INCIBE) and participated in peer code reviews.

Researcher, Lecturer & Teaching Assistant

Faculty of Engineering, Universidad de la República (UdelaR) · Montevideo, Uruguay · 2015 – 2024

- Led independent computational research in statistical modeling and signal processing; designed numerical methods and data-processing pipelines for high-dimensional systems.
- Published **6 peer-reviewed articles** (4 first-author, Scopus-indexed); presented at international conferences (SPIE, IOP, Elsevier).
- Taught university-level physics and computational methods; supervised students in quantitative modeling.

TECHNICAL SKILLS

Machine Learning & Deep Learning

[TensorFlow](#) · [Keras](#) · [PyTorch](#) · [scikit-learn](#) · [MLflow](#) · Unsupervised Learning · Anomaly Detection · Autoencoders / β -VAE · Isolation Forest · Representation Learning · Hyperparameter Optimization · Model Validation · Threshold Calibration · Explainable AI (SHAP/XAI) · Reinforcement Learning

Neuromorphic Computing (current research)

Spiking Neural Networks (SNNs) · Synaptic Plasticity · Event-based Vision (DVS) · SNN-to-FPGA Compilation · Energy-efficient / Edge AI

Data & Infrastructure

[Python](#) (pandas, NumPy, SciPy) · [SQL](#) · [R](#) · [Apache Kafka](#) · [Apache Spark](#) · [Apache Airflow](#) · [MinIO](#) · Streaming architectures · [Git](#) · [Linux](#) · [Docker](#) (basic)

Graph Data Science

[Neo4j](#) (Certified Professional) · [Cypher](#) · Graph Data Modeling · Network Analysis

EDUCATION

- **MSc in Big Data Analytics Technologies** — Universidad de Santiago de Compostela, Spain (2024–2025). Thesis: *Application of new anomaly detection models in Big Data contexts*.
- **Ph.D. in Optics (Physics)** — Universidad de la República, Uruguay (2019–2024). ML coursework: Deep Learning for Computer Vision (CS231n), Reinforcement Learning, Computational Multivariate Statistics, Computer Image Processing.
- **MSc in Theoretical Physics (QCD)** — Universidad de la República, Uruguay (2016–2018).
- **BSc in Physics** — Universidad de la República, Uruguay (2011–2015).

CERTIFICATIONS

- **Neo4j Certified Professional** — Neo4j GraphAcademy, May 2026 ([full profile](#))
- **Master Python for DevOps and CI/CD** — Udemy, 2025

SELECTED PUBLICATIONS

- **[2024 - SPIE]** [Integration with one partial derivative applied to quantitative phase imaging](#)
- **[2023 - Optics and Lasers in Engineering]** [Common-path quantitative phase imaging by propagation through a sinusoidal intensity mask](#)

6 peer-reviewed publications in total (4 first-author, Scopus) — full list on [ORCID](#).

LANGUAGES

Spanish — Native · **English** — Professional working proficiency

Last updated: July 2026